

1. Two bodies at different ^W temperatures are placed in thermal contact,

(a) Assuming no interaction with the surroundings, which of the following is/are true?

(i) Each system is isolated; (ii) One of the systems can be isolated, but the other one is not;

~~(iii)~~ the composite system made of both A and B together is isolated.

(b) If in a process, the changes in entropy for the two bodies are respectively, ΔS_A and ΔS_B , then is $\Delta S = \Delta S_A + \Delta S_B$ must be (choose one) ' $>$ ', ' $=$ ' or ' $<$ ' 0

(c) Suppose the initial temperatures of the bodies are respectively T_A and T_B . If heat capacities, $C_{A(B)}$ remain constant, for max. work extracted in processes occurring, find ΔS_A , if the final temperature in the composite system is T .

(d) Find T in terms of $C_{A(B)}$ and $T_{A(B)}$

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2. What does isothermal compressibility, κ_T , mean physically? Show that for a van der Waals gas, $\kappa_T = \frac{V^2(V-b)^2}{RTV^3-2a(V-b)^2}$. How
2 + 2 + 2
 and why is it different from the expression for a perfect gas?

3. The standard reaction enthalpy for hydrogenation of propene, $\text{CH}_2=\text{CHCH}_3(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{CH}_3\text{CH}_2\text{CH}_3(\text{g})$ is -124 kJ mol^{-1} .
The standard reaction enthalpy for combustion of propane, $\text{CH}_3\text{CH}_2\text{CH}_3(\text{g}) + 5 \text{ O}_2(\text{g}) \rightarrow 3 \text{ CO}_2(\text{g}) + 4 \text{ H}_2\text{O}(\text{l})$ is $-2220 \text{ kJ mol}^{-1}$. The standard reaction enthalpy for formation of water, $\text{H}_2(\text{g}) + \frac{1}{2} \text{ O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ is -286 kJ mol^{-1} . Calculate the standard enthalpy of combustion of propene.

4. Draw a S versus p plot for the Carnot cycle. Label the axes properly. Write expressions for ΔS for each step.

$$3 + 2$$

$$\Delta S_{\text{net}}$$

5. Consider a function $\xi(x, y, z)$ and a relation exists among x , y and z . Use $x \equiv x(\xi, y)$ to expand dx and $y \equiv y(\xi, z)$ to expand dy . Use the two equations to (i) obtain the value of $\left(\frac{\partial x}{\partial y}\right)_\xi \left(\frac{\partial y}{\partial z}\right)_\xi \left(\frac{\partial z}{\partial x}\right)_\xi$ and (ii) show $\left(\frac{\partial x}{\partial \xi}\right)_z = \left(\frac{\partial x}{\partial \xi}\right)_y + \left(\frac{\partial x}{\partial y}\right)_\xi \left(\frac{\partial y}{\partial \xi}\right)_z$ 3 + 3

6. Energy dissipation resulting from a nanomagnetic bit flipping from off to on requires ≈ 0.026 eV (4.2×10^{-21} J) at 300 K. By how much (percentage) is it lower or higher than the Landauer minimum ($k_B = 1.38 \times 10^{-23}$ J/particle)?

Bonus : A line in the Flanders & Swann thermodynamics song is 'Heat is work and work is heat'. Is this a correct statement?

Comment in less than 50 words.

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